

1. Industrial Application & Purpose

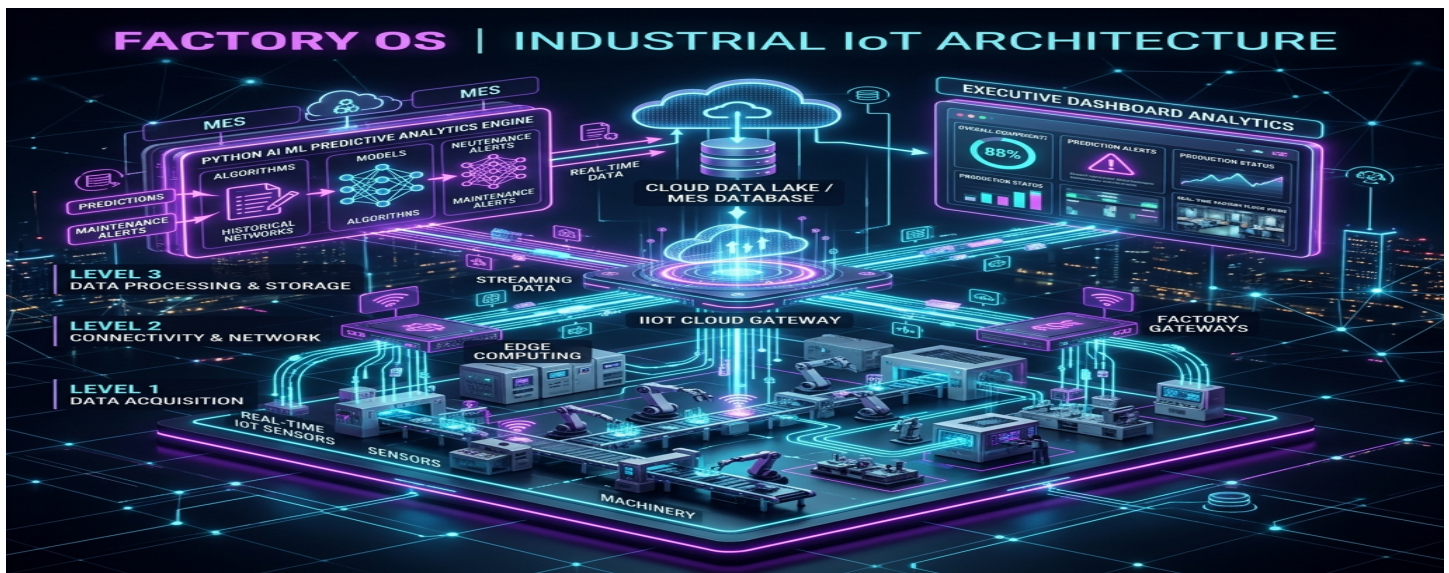
What is the Use of this App to Industries?

Factory OS is a next-generation Manufacturing Execution System (MES) and Industrial IoT Decision Intelligence Platform built for high-speed assembly environments.

* Core Purpose: Enables plant managers, reliability engineers, and production supervisors to monitor multi-site telemetry in real-time, predict equipment failures before breakdown, and enforce >95.0% Overall Equipment Effectiveness (OEE) compliance.

* Target Industries: Automotive Giga-Assembly, Battery Module Manufacturing, Heavy Hydraulic Stamping, and Electronics Line Optimization across global sites (Detroit, Austin, Berlin, Shanghai).

* SAP ERP Integration: Directly connects to SAP PM (Plant Maintenance) and SAP MM (Materials Management) BAPI endpoints for automated work orders and inventory refills.



2. Telemetry Data Scale & Volumes

How Much Data is Used?

Factory OS processes high-throughput industrial telemetry streams across 20 registered industrial machines:

- * IoT Sensor Sampling Rate: 100 Hz High-Speed Streaming (100 readings per second per machine).
- * Daily Log Volume: Over 25 Million sensor data points daily (Operating Speed, Hydraulic Pressure, Thermal Drift, ISO 10816 Vibration).
- * Historical Vault: 30-Day Rolling Time-Series history per plant location.
- * Stochastic Monte Carlo Scale: 1,000 trial parameter iterations computed per simulation execution.

Telemetry Channel	Sampling Rate	Daily Data Volume	Retention Vault
Operating Speed (SPM)	100 Hz	8.64 Million Records	30 Days Time-Series
Hydraulic Pressure (Bar)	100 Hz	8.64 Million Records	30 Days Time-Series
Core Temperature (C)	10 Hz	864,000 Records	90 Days Historical
ISO 10816 Vibration (mm/s)	100 Hz	8.64 Million Records	30 Days Time-Series

3. Python 3.13 AI/ML Algorithms

What Algorithms are Used & How do They Run?

1. Multi-Variable Z-Score Anomaly Engine:

$$\text{Formula: } Z = \sqrt{(Z_{\text{speed}}^2 + Z_{\text{press}}^2 + Z_{\text{temp}}^2 + Z_{\text{vib}}^2) / 4}$$

Evaluates deviation across 4 sensor channels. Triggers alert if $Z > 2.25$.

2. Logistic Sigmoid Failure Risk Classifier:

$$\text{Formula: } P(\text{Failure}) = 100 / (1 + e^{-\text{logit}})$$

Calculates failure probability % based on continuous mechanical stress.

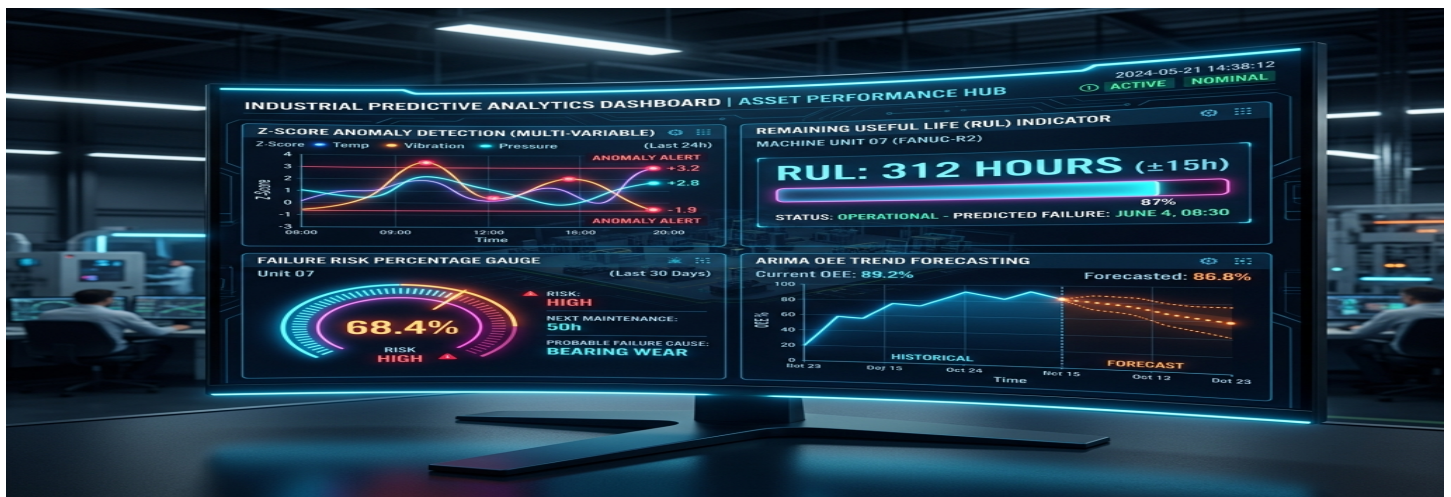
3. Remaining Useful Life (RUL) Regressor:

$$\text{Formula: } \text{RUL} = \max(0, 168 / \text{degradation_index})$$

Estimates remaining operational hours before mandatory maintenance.

4. ARIMA(1,1,0) & Monte Carlo (1,000 Trials):

Extends 24-hour OEE curves into a 6-shift forecast & computes $P(\text{OEE} \geq 95\%)$ confidence limits.



4. Industrial Problems Solved

What Problem is Solved by this App?

1. Zero Unplanned Mechanical Downtime:
Replaces reactive breakdown downtime (\$22,000/min cost) with 140-hour advance RUL predictive failure alerts.
2. Six Sigma Quality Control (<142.5 DPMO):
Cognex Vision AI detects micro-weld voids automatically, maintaining 98.8% First Pass Yield (FPY).
3. Automated ERP Work Orders & Logistics:
Dispatches SAP PM tickets and SAP MM purchase requisitions when stock drops below 25% safety threshold.

